Sleep Study on the Job

An investigation into sleep qualities and occupation.

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A deep dive into the correlation between sleep quality and occupation

Importance and Context

In our study we aim to explore potential connections between sleep quality and occupation using a dataset containing occupations, sleep duration, stress levels, BMI, blood pressure, daily steps, gender, age, physical activity, and finally sleep quality. Our primary question is: "Is there a relationship between sleep quality and different occupations?" This is relevant because some jobs are may be more demanding or stressful, potentially impacting sleep patterns. Previous studies have shown mixed results, examining short sleep prevalence, working hours, and occupational stress. We will use linear regression models and plots to detect trends and statistically significant relationships, aiming to uncover insights about how different careers may influence sleep.

Data and Methodology

In our research, we utilized a data set containing sleep and health/lifestyle data with 374 individual unique Person IDs. We have narrowed this information down to exclude occupations that we deem to contain too few entrants. We set the cutoff for this at 20, meaning, we removed occupations that made up around 5% or less of the population. This helped to ensure more accuracy in our models, and brought us down to 7 Occupations out of the original 10. This left us with a total of 363 rows of data, which we deemed sufficient for this study.Each remaining row contains multiple observations, collected at once from each individual. We specifically would like to focus on the Y-variable of sleep quality, as we felt we may see unexpected responses when compared to sleep duration that could create notable diversity between our occupations. Sleep quality is a 1-10 **Y** measurement that we will be treating as a continuous variable in this case. We will also be comparing captured sleep quality to our other X variables in our dataset as we explore.

Here, we show a model giving a visual of our coefficients for quality of sleep based on occupation.

```
Call:
lm(formula = Quality.of.Sleep ~ Occupation, data = exploration_set)

Residuals:
     Min       1Q   Median       3Q      Max
-2.09524 -0.44444  0.07143  0.19048  2.19048

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)            7.32431    0.08694  84.241  < 2e-16 ***
OccupationAccountant   0.58478    0.23441   2.495  0.01426 *
```

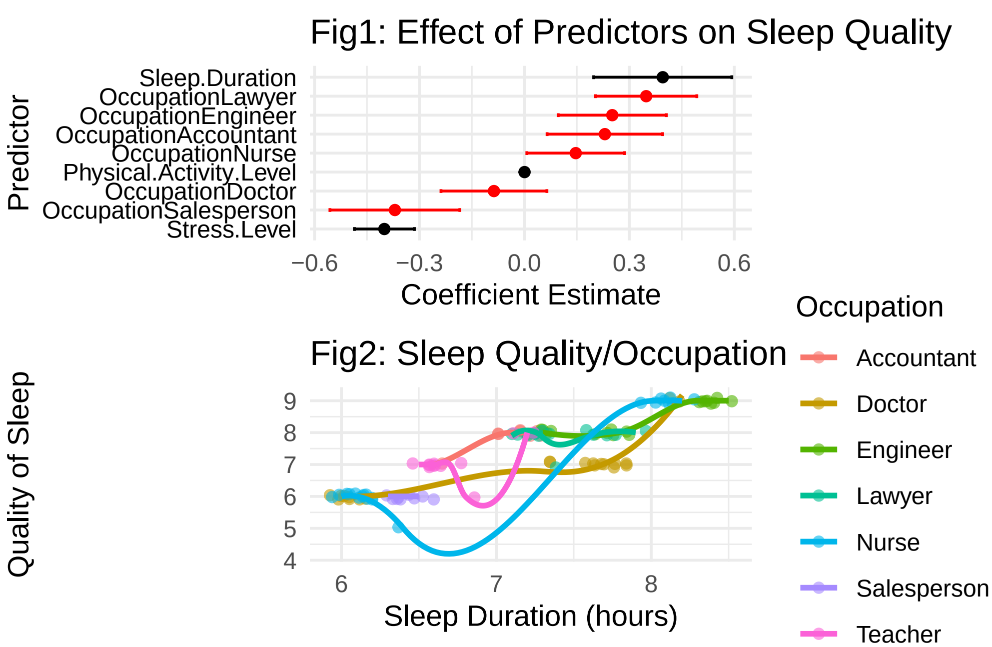
```
OccupationDoctor      -0.51479    0.17995  -2.861  0.00516 **
OccupationEngineer     1.12013    0.19110   5.862 6.04e-08 ***
OccupationLawyer       0.60426    0.21164   2.855  0.00524 **
OccupationNurse       -0.22908    0.17995  -1.273  0.20599
OccupationSalesperson -1.32431    0.25588  -5.175 1.19e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8543 on 99 degrees of freedom
Multiple R-squared:  0.4296,    Adjusted R-squared:  0.395
F-statistic: 12.43 on 6 and 99 DF,  p-value: 2.164e-10
```

From this summary of our exploratory sleep quality model, we made some important tweaks. As we do not have a baseline occupation, we instead took a grand mean to use as our baseline, incorporating all occupations. This number comes to an average of 7.324 quality of sleep. Based off of this data alone, we see a significant p-value for Accountant(+0.58, $p < 0.05$), Doctor(-0.51, $p < 0.01$), and Lawyer(+0.60, $p < 0.01$), very strong significance in both Engineer(+1.12, $p < 0.001$), and Salesperson(–1.32, $p < 0.001$), and a value without statistical significance in Nurses. With their coefficients in mind, we can mixed, yet significant results. Our R squared gives us a 43% proportion of variance in sleep quality and an adjusted value of 39.5% explained by occupation by this slice of data and our number of predictors. With our F-statistic and p-value, we can convincingly imagine that in this data slice, Occupation is a significant predictor of quality of sleep.

```
Call:
lm(formula = Quality.of.Sleep ~ Sleep.Duration + Occupation +
    Stress.Level + Physical.Activity.Level, data = exploration_set)

Residuals:
     Min       1Q   Median       3Q      Max
-1.51769 -0.12771 -0.01268  0.15407  0.67492

Coefficients:
                         Estimate Std. Error t value Pr(>|t|)
(Intercept)             6.6297570  0.8771469   7.558 2.42e-11 ***
Sleep.Duration          0.3954567  0.0994568   3.976 0.000136 ***
OccupationAccountant    0.2299514  0.0832190   2.763 0.006861 **
OccupationDoctor       -0.0871412  0.0762434  -1.143 0.255908
OccupationEngineer      0.2511225  0.0778961   3.224 0.001729 **
OccupationLawyer        0.3481317  0.0728712   4.777 6.38e-06 ***
OccupationNurse         0.1467481  0.0704177   2.084 0.039819 *
```

```
OccupationSalesperson   -0.3702611  0.0935159  -3.959 0.000144 ***
Stress.Level            -0.4003904  0.0431797  -9.273 5.44e-15 ***
Physical.Activity.Level  0.0003713  0.0017578   0.211 0.833160
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2792 on 96 degrees of freedom
Multiple R-squared:  0.9409,    Adjusted R-squared:  0.9354
F-statistic: 169.8 on 9 and 96 DF,  p-value: < 2.2e-16
```

With the addition of Stress Level, Physical Activity Level, and Sleep Duration, our model now explains 94.09% of sleep quality variance. We chose to omit age and gender here, and even the existence of sleep disorders. This slice shows that sleep duration (+0.40, $p < 0.001$) and stress level(–0.40, $p < 0.001$) contribute heavily to sleep quality,though physical activity does not appear significant here. There also seems to be a shift in the effect of occupation on sleep quality, with Doctor no longer appearing significant, and appearing to have a higher tendency to stress and/or have lower sleep duration independently. These could obviously be overlapping with occupation, but their addition enhances our model nonetheless.



Here is a visual(Figure 1 or Fig1) of our X-variables effects (coefficient) on our outcome variable. We see that sleep duration and stress level are most significant, but when directly evaluating for occupation (in red), Salesperson has a very significant negative correlation, and Lawyer as a very significant positive correlation. We can see in Figure 2 (or Fig2), that there is in fact, a positive trend with sleep duration and sleep quality as expected, with some occupations clearly falling behind regardless. We use jittered points here to give a stronger representation of our data visually, and give a properly continuous look.

Results

With our confirmation set, we will test how well our exploration set can be used to predict our values.

Predicted \ Actual	1_QoS	2_QoS	3_QoS	4_QoS	5_QoS	6_QoS	7_QoS	8_QoS	9_QoS	10_QoS
1_QoS	0	0	0	0	0	0	0	0	0	0
2_QoS	0	0	0	0	0	0	0	0	0	0
3_QoS	0	0	0	0	0	0	0	0	0	0
4_QoS	0	0	0	0	0	0	0	0	0	0
5_QoS	0	0	0	0	4	3	0	0	0	0
6_QoS	0	0	0	0	0	67	0	0	0	0
7_QoS	0	0	0	0	1	2	45	0	0	0
8_QoS	0	0	0	0	0	0	7	72	0	0
9_QoS	0	0	0	0	0	0	2	3	50	0
10_QoS	0	0	0	0	0	0	0	0	0	0

We can see here, that Quality of Sleep (QoS) also has strong predictability. This is a very good sign that our data is significant due to accurate predictions of sleep quality values using our exploration set.

Discussion

Our analysis found a meaningful relationship between occupation and self-reported sleep quality. While job title alone explained some variation, our stronger model, adjusting for sleep duration and stress offered a clearer picture. Sleep duration and stress level were the most influential predictors, but occupation remained an important factor. Notably, Salespeople consistently reported significantly lower sleep quality across both models. This suggests that unmeasured factors such as irregular hours, travel, or performance pressure may uniquely impact sleep in this group. Future research could explore these dimensions more closely. Additionally, it is important to note that the data is entirely self-reported. Differences in how individuals perceive and report sleep and stress may vary by occupation, introducing bias. Incorporating objective measures in future studies would help validate these findings. In sum, while stress and sleep duration strongly influence sleep quality and likely have occupational overlap, occupation itself continues to matter, suggesting it's a valuable lens for understanding sleep health in the workforce.

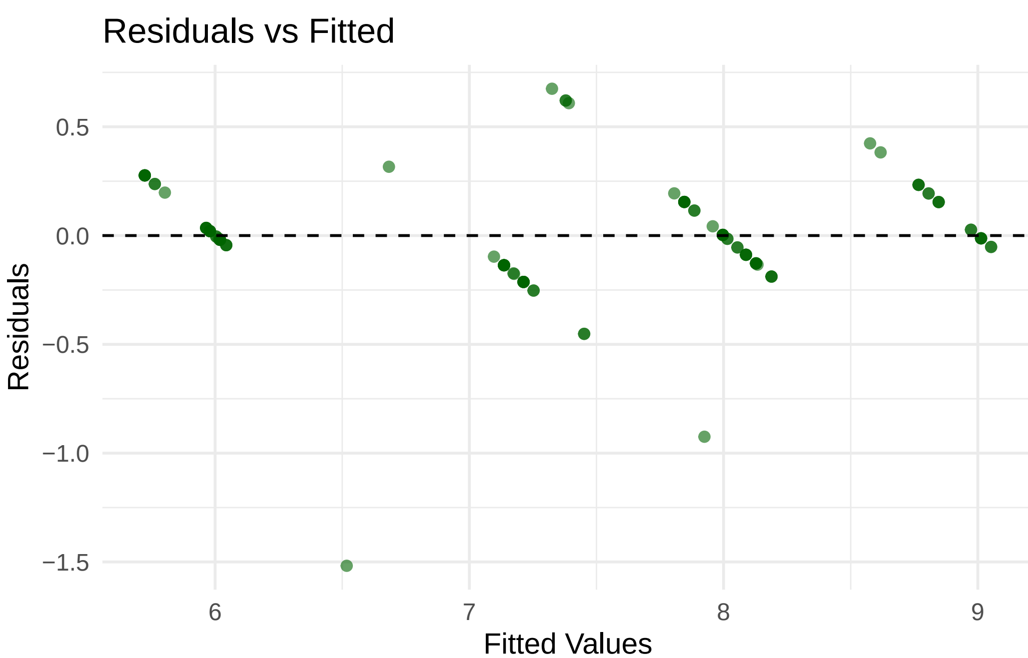
Appendix

https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset

For a first model we chose Occupation and Sleep quality due to the fact that these are the two variables we are most interested in seeing the relationship between, and the core of the study.

For a second model we used sleep duration, sleep quality, occupation, and stress level because we believed that stress level and sleep duration could potentially be more effective in predicting sleep quality than occupation, making it a reliable way to test our hypothesis.

A third model was also created, showing high sleep quality predictive value on sleep duration and stress level, but due to the focus of our report being occupation, it was not included



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